Quiz 10: Solutions

Give exact answers and justify your reasoning. Do not use electronic tools. All functions are real-valued, and angles are measured in radians. Local extrema are considered only at interior points of the stated domain. Use the strict tangent-line definition of concavity from Lecture 11.

1. Determine the maximal open intervals of concavity and all inflection points of $f(x) = \ln(1 + x^2)$. Give the coordinates of each inflection point.

Solution.

The function is defined and twice differentiable on $\mathbb{R}$, with

$$f'(x) = \frac{2x}{1 + x^2}, \qquad f''(x) = \frac{2(1 + x^2) - 4x^2}{(1 + x^2)^2} = \frac{2(1 - x^2)}{(1 + x^2)^2}.$$

The denominator is positive for every x. Thus $f''(x) > 0$ when $-1 < x < 1$, and $f''(x) < 0$ when $x < -1$ or $x > 1$. The maximal open intervals are therefore

$$\begin{aligned} \text{concave up:} &\quad (-1, 1), \\ \text{concave down:} &\quad (-\infty, -1) \text{ and } (1, \infty). \end{aligned}$$

The function is continuous at -1 and 1, and its concavity changes at both points. Hence the inflection points are

$$\boxed{(-1, \ln 2) \quad \text{and} \quad (1, \ln 2).}$$

There are no others, since the concavity is constant on each of the three intervals above.

2. Use the Mean Value Theorem to prove

$$|\sin x - \sin y| \le |x - y| \qquad (x, y \in \mathbb{R}).$$

Deduce an interval with rational endpoints containing $\sin(\pi/6 + 1/20)$, without using decimal approximations.

Solution.

If $x = y$, both sides of the required inequality are zero. If $x \neq y$, sine is continuous on the closed interval with endpoints x and y and differentiable on its interior. The Mean Value Theorem gives a number c strictly between them such that

$$\sin x - \sin y = \cos c\,(x - y).$$

Since $|\cos c| \le 1$,

$$\boxed{|\sin x - \sin y| \le |x - y|.}$$

Now take $x = \pi/6 + 1/20$ and $y = \pi/6$. Using $\sin(\pi/6) = 1/2$, we obtain

$$\left|\sin\left(\frac{\pi}{6} + \frac{1}{20}\right) - \frac{1}{2}\right| \le \frac{1}{20}.$$

Thus an interval with rational endpoints containing the required value is

$$\boxed{\left[\frac{9}{20}, \frac{11}{20}\right].}$$